Bringing Back the Context: Camera Trap Species Identification as Link Prediction on Multimodal Knowledge Graphs

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Camera Trap Wildlife Recognition Challenges



Illumination: Animals are not always salient

Image Credit: Beery et al. 2020

Camera Trap Wildlife Recognition Challenges





Motion blur: common with poor illumination at night

Image Credit: Beery et al. 2020

Camera Trap Wildlife Recognition Challenges







Size of the region of interest (ROI): Animals can be small or far from the camera

Image Credit: Beery et al. 2020

Camera Trap Wildlife Recognition Challenges









Occlusion: e.g. by bushes or rocks

Image Credit: Beery et al. 2020

Camera Trap Wildlife Recognition Challenges











Camouflage: decreases saliency in animals' natural habitat

Image Credit: Beery et al. 2020

Camera Trap Wildlife Recognition Challenges







Perspective: Animals can be close to the camera, resulting in partial views of the body.





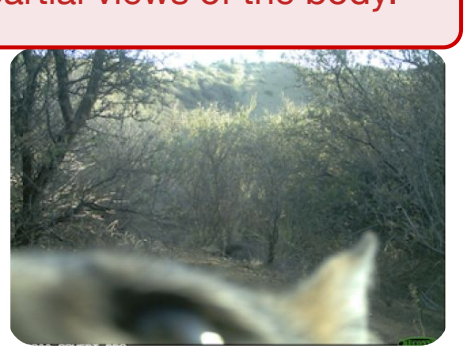


Image Credit: Beery et al. 2020

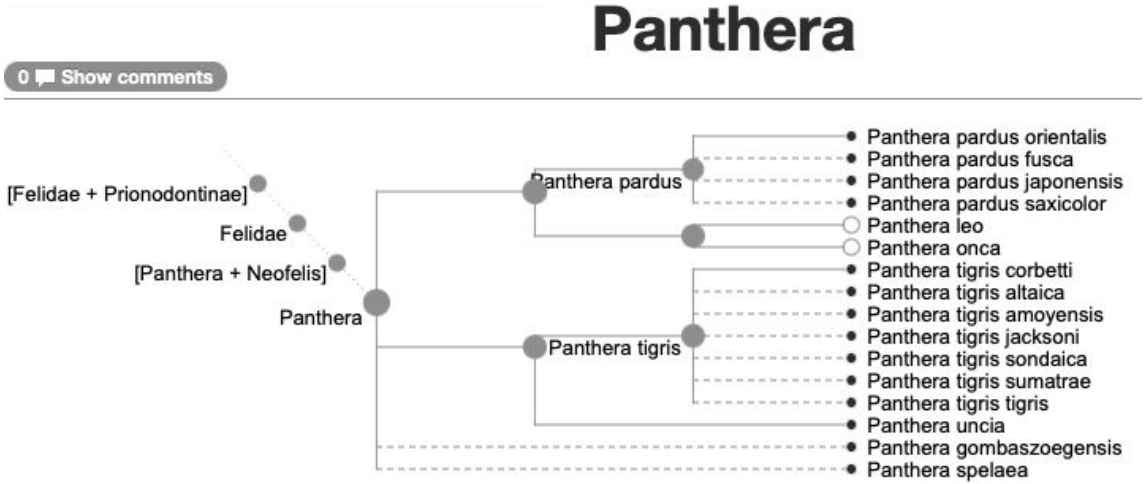
Camera Trap Wildlife Recognition Challenges

Train			Test (OOD)
d = Location 1	d = Location 2	d = Location 245	d = Location 246
			
Vulturine Guineafowl	African Bush Elephant	unknown	Wild Horse
			
Cow	Cow	Southern Pig-Tailed Macaque	Great Curassow

Figure Credit: Koh et al. 2021

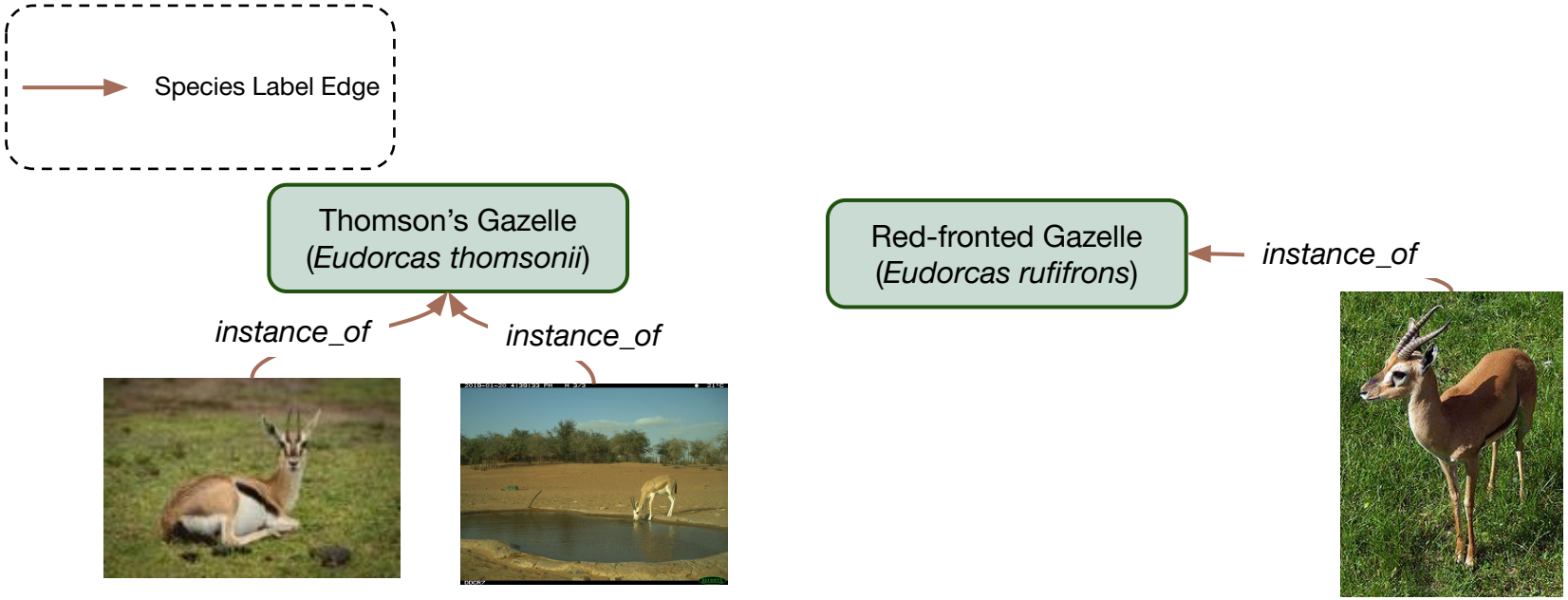
Wildlife Taxonomy

- Phylogenetic taxonomy from Open Tree of Life (OTT)



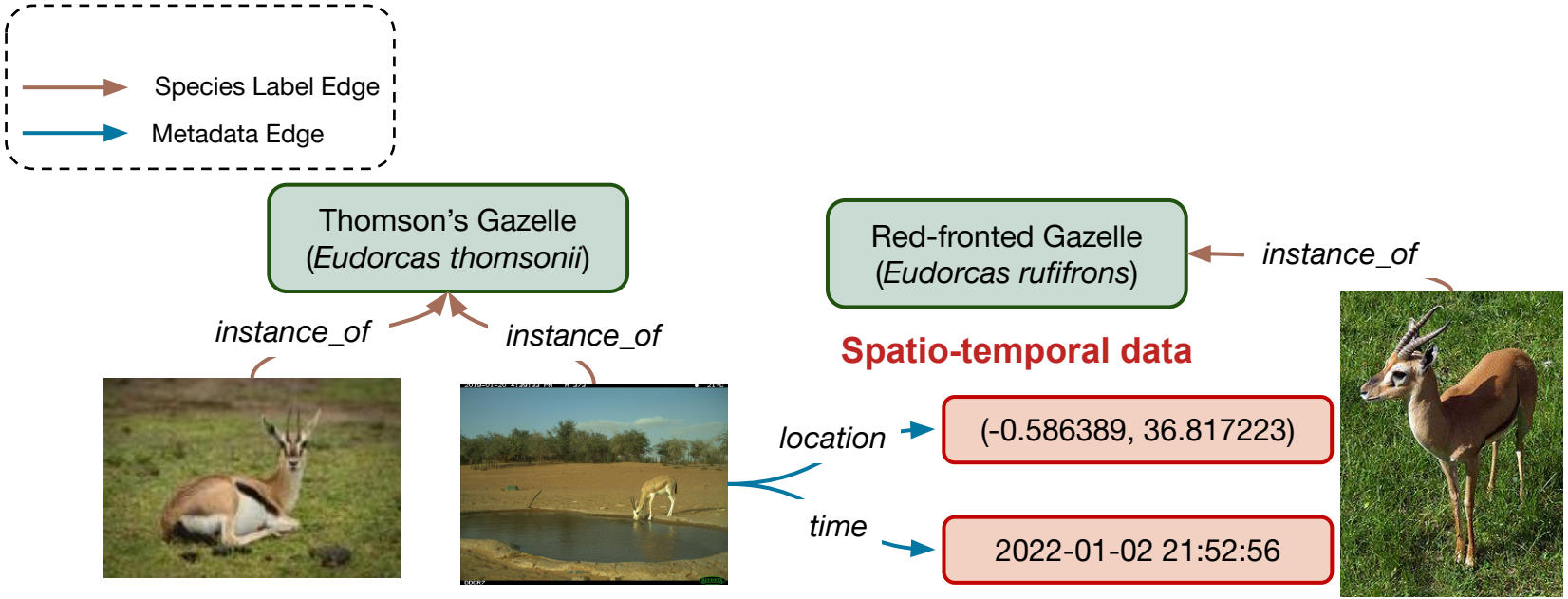
Multimodal KG

- Structured representation for semantic relationships b/w wildlife species, images, attributes and taxonomy.



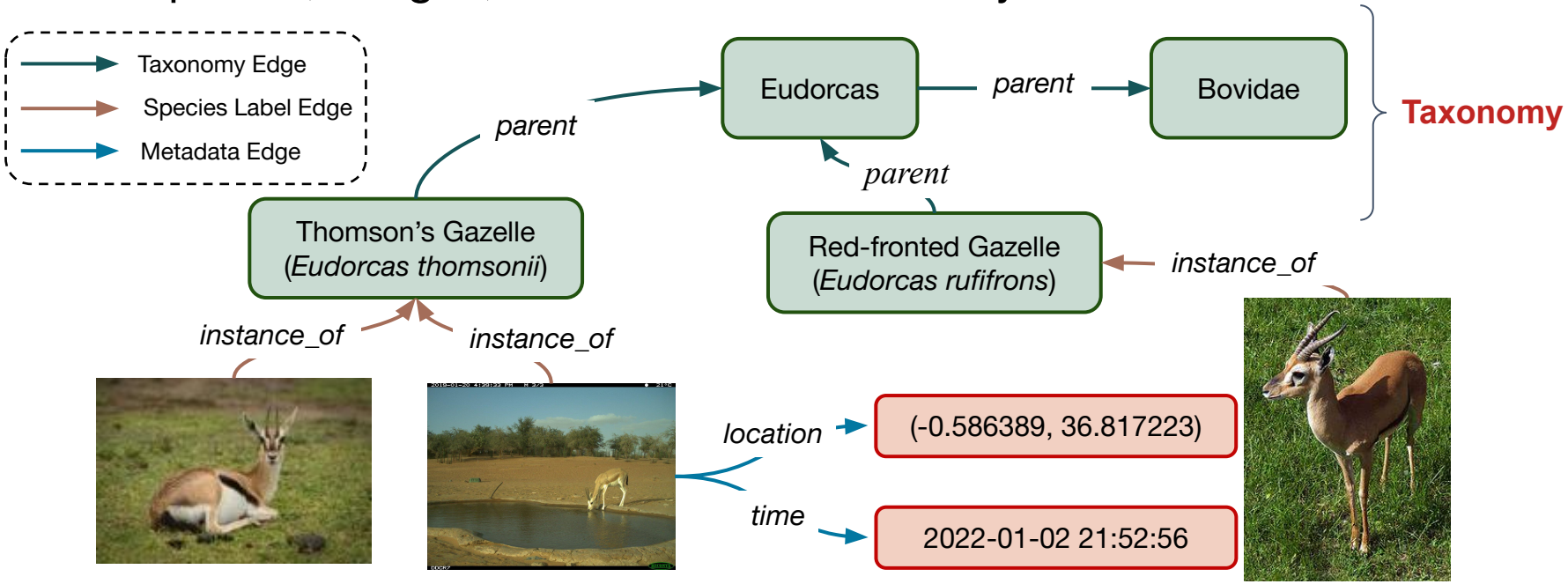
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Multimodal KG

- Entity types:
  - Categorical (e.g. Species labels, taxons)
    - Encoder: Entity lookup table

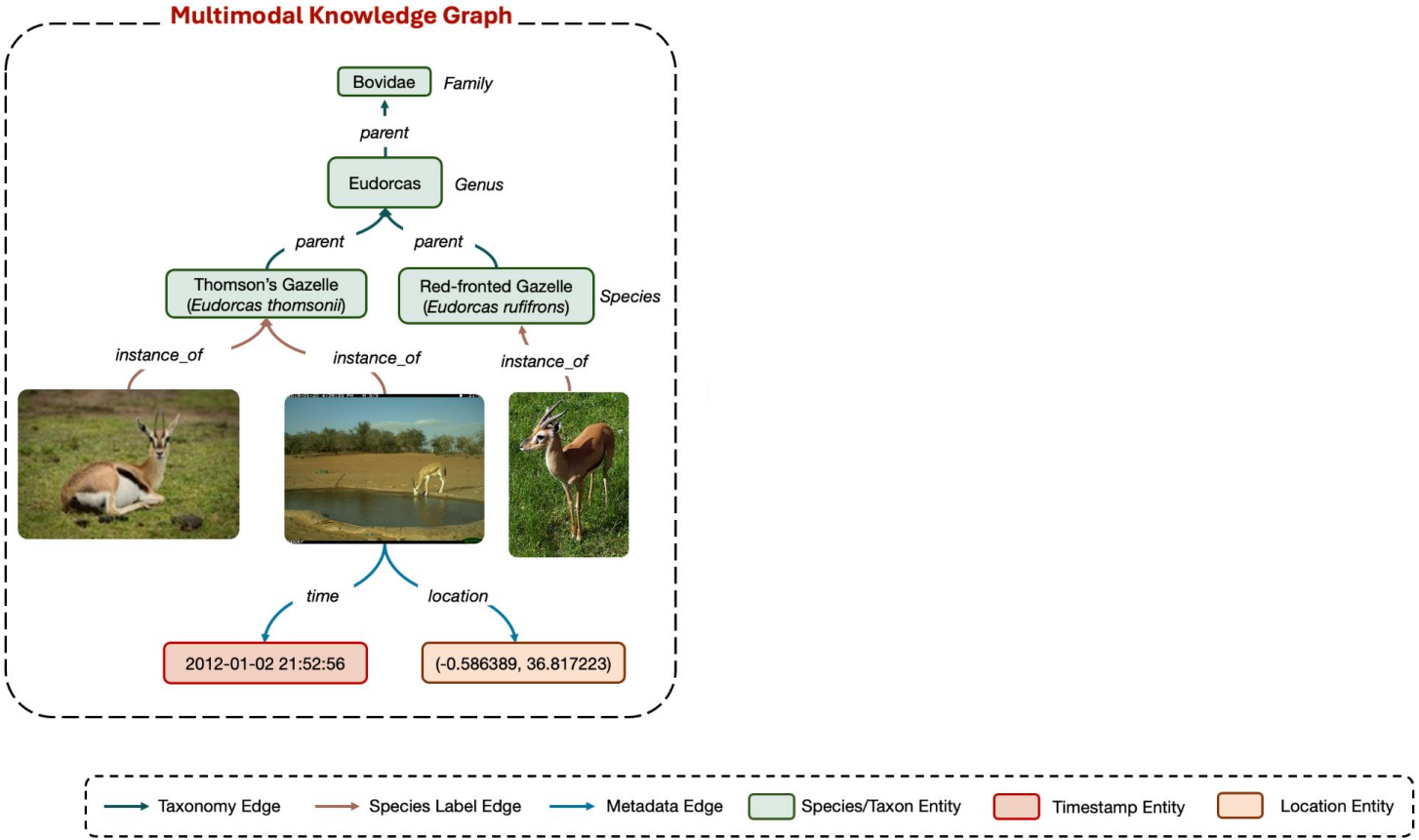
Multimodal KG

- Entity types:
  - Categorical (e.g. Species labels, taxons)
    - Encoder: Entity lookup table
  - Visual
    - Encoder: CNN/Vision Transformer

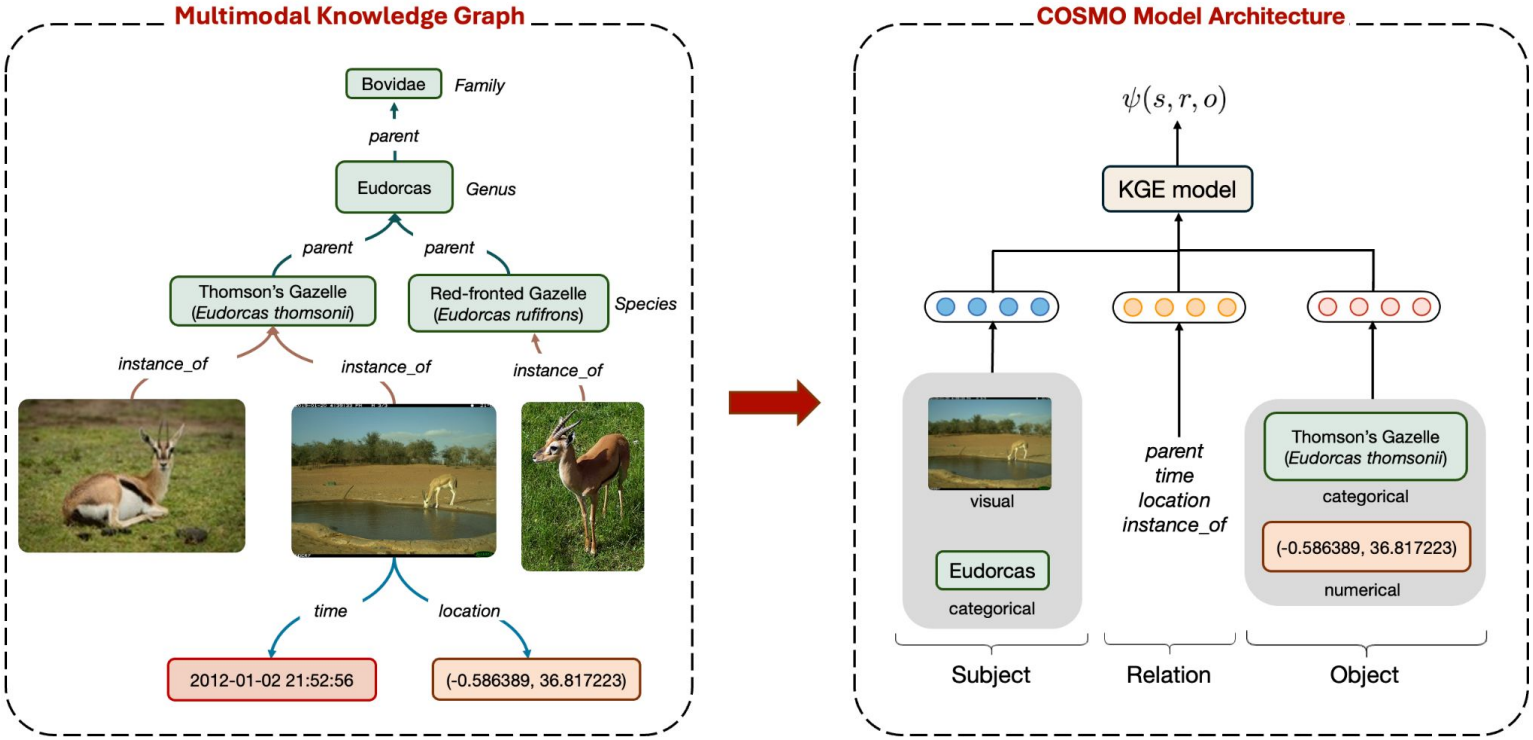
Multimodal KG

- Entity types:
  - Categorical (e.g. Species labels, taxons)
    - Encoder: Entity lookup table
  - Visual
    - Encoder: CNN/Vision Transformer
  - Numerical (e.g. location, time)
    - Encoder: MLP

Multimodal KG Embedding Model



Multimodal KG Embedding Model



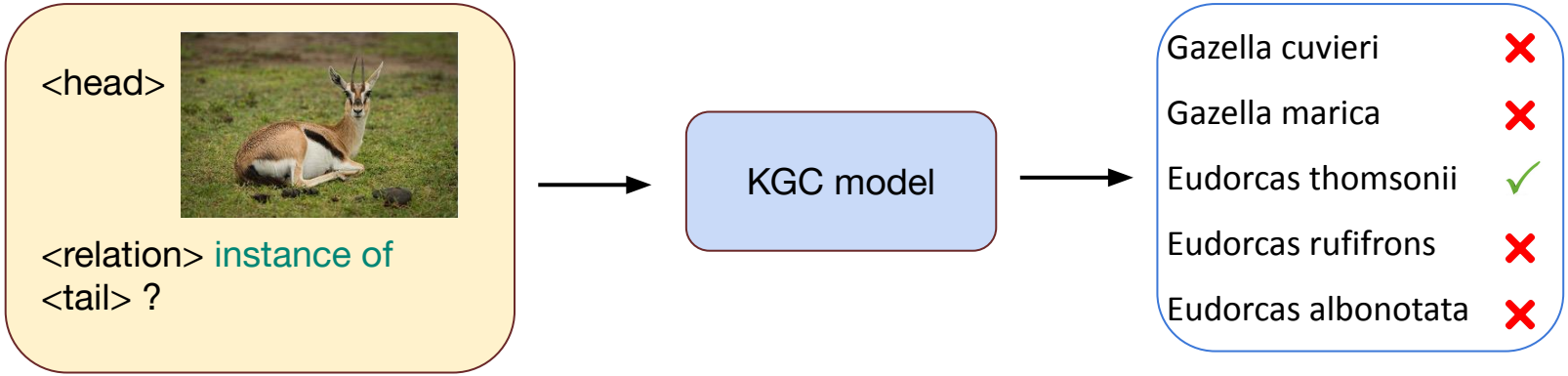
Taxonomy Edge | Species Label Edge | Metadata Edge | Species/Taxon Entity | Timestamp Entity | Location Entity

Task: Species Classification

- Task: Given a camera trap image, predict the species label.

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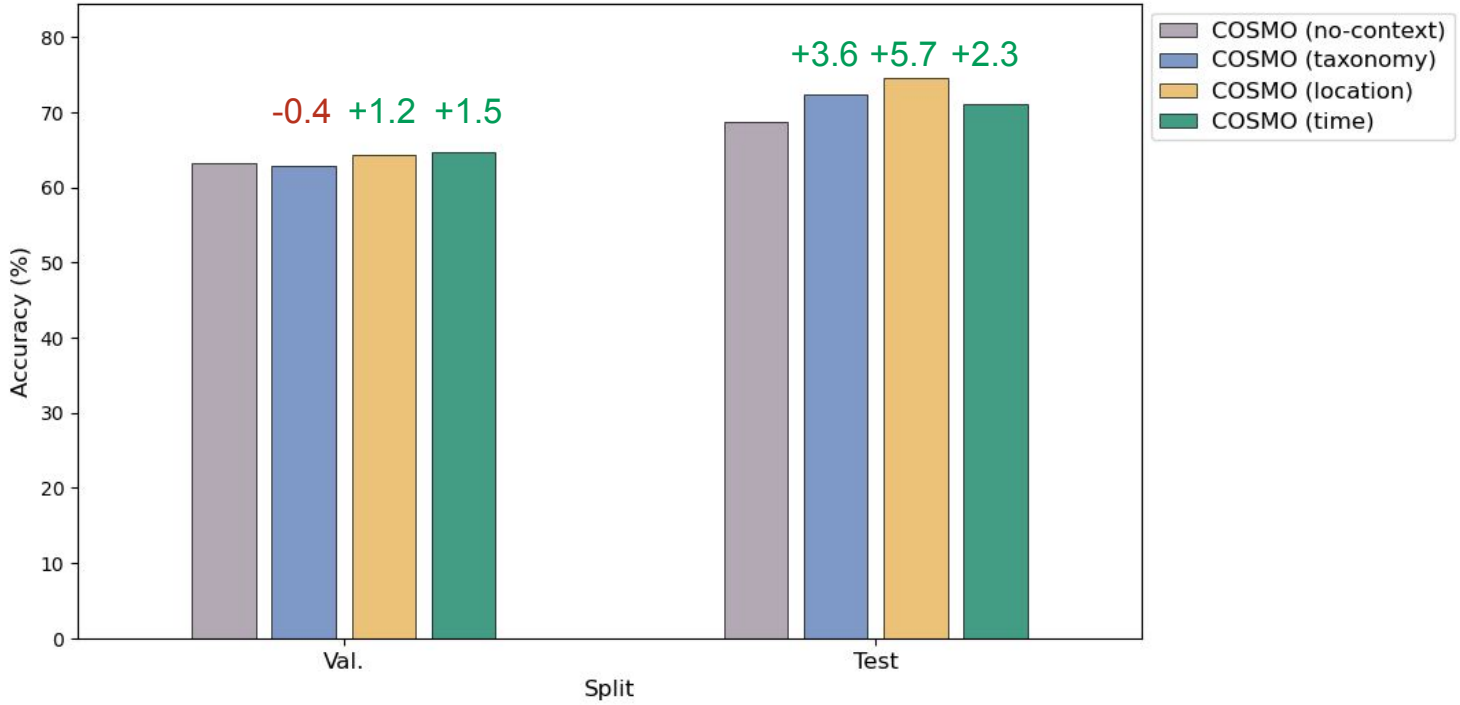
Input: Structured Query

Output: Candidate species with scores

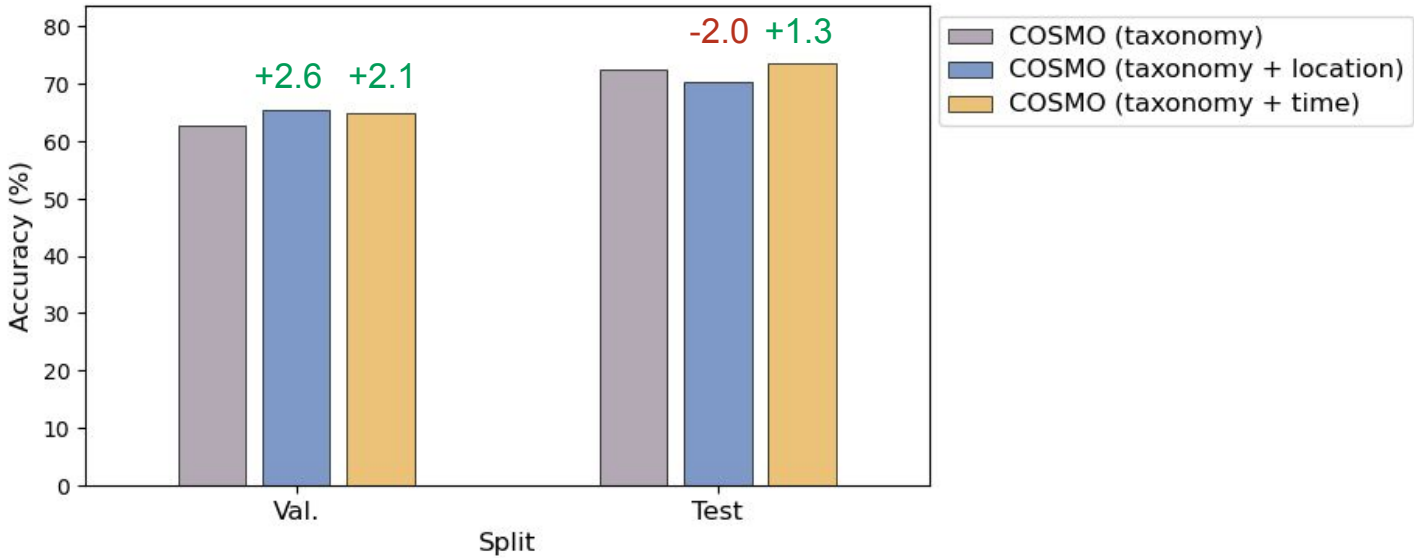
Training

- Categorical attributes: Multi-class classification problem
- Numerical attributes: Multi-class multi-label classification problem.
- Sequentially train on each set of triples types.

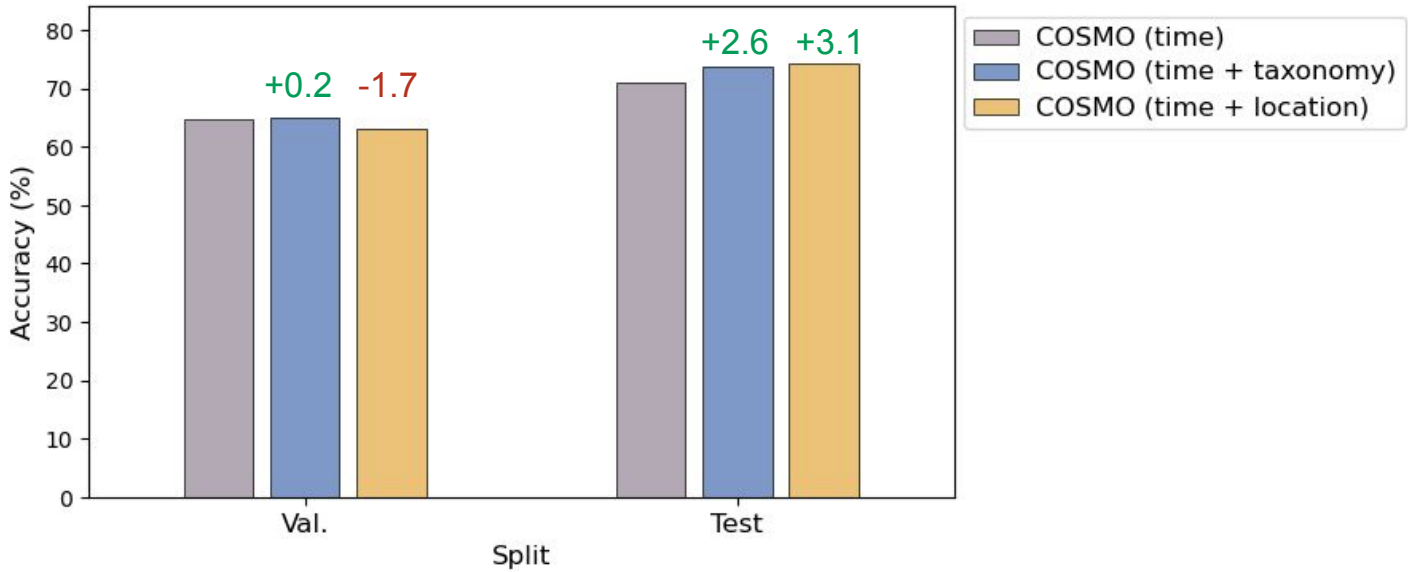
Results


COSMO (no-context)
COSMO (taxonomy)
COSMO (location)
COSMO (time)
-0.4
+1.2
+1.5
+3.6
+5.7
+2.3
Accuracy (%)
0
10
20
30
40
50
60
70
80
Val.
Test
Split

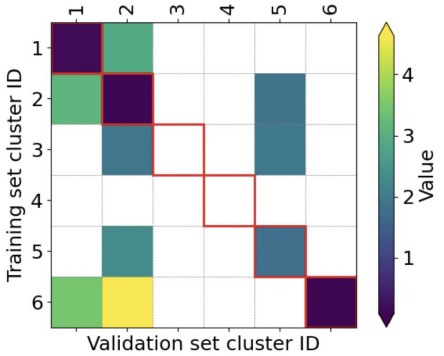
Results (Multiple contexts)



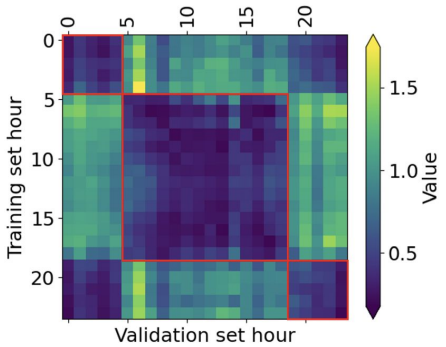
Results (Multiple contexts)



Metadata Analysis



(a) Each color square shows the distance between the corresponding validation cluster centroid on x-axis and the training cluster centroid on y-axis. The correlation peaks along the diagonal (highlighted in red)[3].



(b) Each color square shows the distance between the corresponding training hour slot on x-axis and validation hour slot on y-axis. The correlation peaks for day-day and night-night hour slots (highlighted in red).

- Similarity peaks along the diagonal for the location attribute, as well as for the day/night categorization of the time attribute.
- These metadata give a prior for species class distribution

Key Takeaways

- Unified way to leverage various forms of multimodal context for species classification in camera traps.
- Our framework achieves superior out-of-distribution generalization and competitive performance with state-of-the-art.

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Contact: pahuja.9@osu.edu

Code: https://github.com/OSU-NLP-Group/COSMO

Thank You!